

IV. Results — Model Performance Comparison

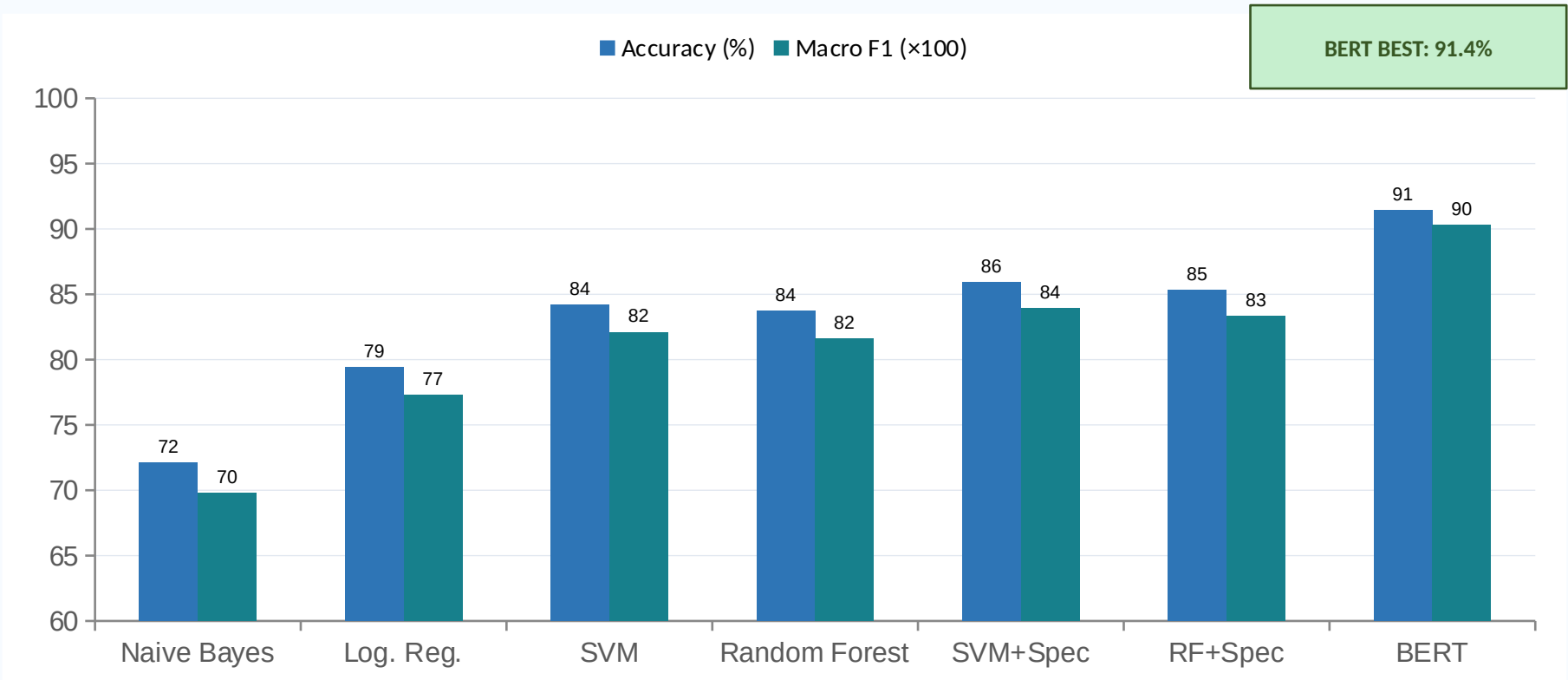


Fig. 4. Grouped bar chart comparing classification accuracy and macro F1-score (×100) across all seven evaluated model configurations on the five-class sentiment task.

IV. Results — Per-Class F1 Score Analysis

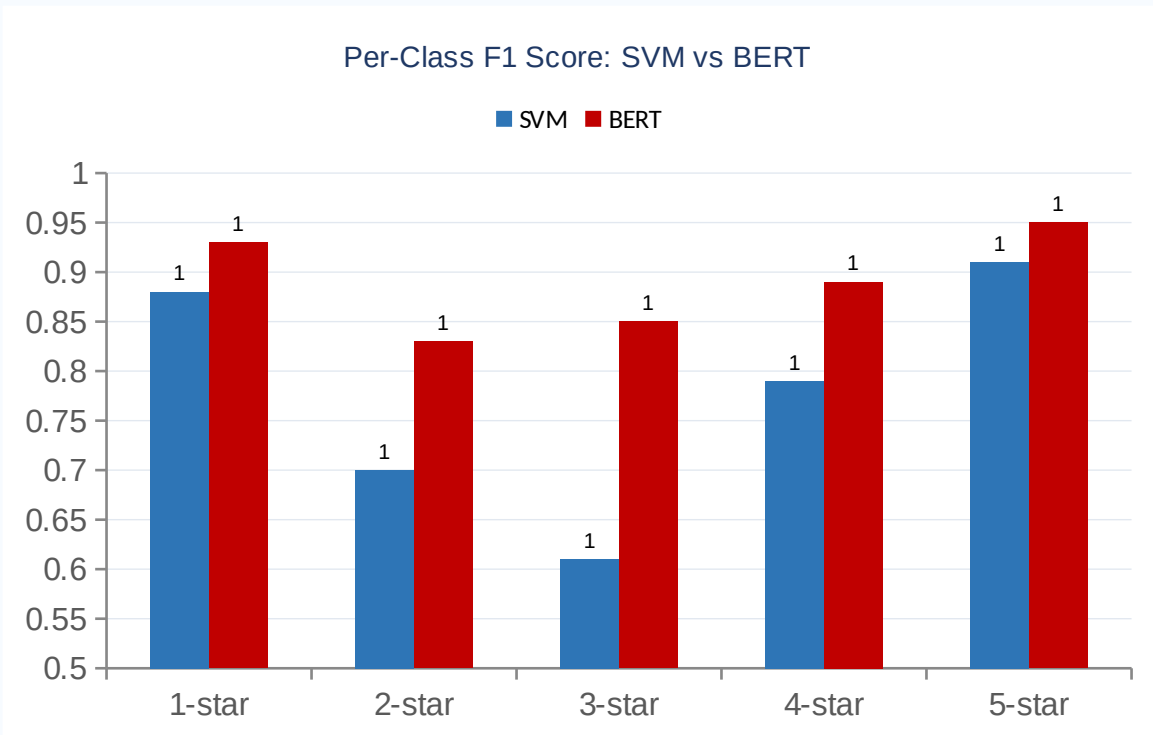


Fig. 5. Per-class F1-score comparison between SVM (best classical) and fine-tuned BERT on all five rating categories. BERT outperforms SVM most significantly on the neutral 3-star class (+23.5 percentage points).

KEY FINDINGS

BERT 3-star F1

0.847 vs 0.612 (SVM)

BERT Overall Acc.

91.4% +7.2pp over SVM

Class Imbalance Fix

+4.1pp Weighted vs unweighted F1

Spec Features Gain

+1.7pp SVM with spec features